I am still confused about the actual data for ECG leads in an ISHNE file: I cannot tell if the data is stored as 16-bit integer value or as a float. Can you make sense of it?

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You are right to be confused, as the answer isn't one or the other. The raw data in an ISHNE file is stored as a **16-bit signed integer** [4]. However, this integer is a raw measurement from the Analog-to-Digital Converter (ADC). To get the actual voltage value, you must convert it to a physical unit (like microvolts) using the **amplitude resolution** value defined in the file's header [4].

Here's a breakdown of the data storage:

📊 Data Type and Format

- **Stored As:** Each ECG sample for every lead is stored as a **2-byte (16-bit) signed integer**, which can represent values from -32,768 to 32,767 [4].
- **Sample Order:** The samples are stored in a multiplexed binary format, meaning they are arranged by sample number. For a 12-lead recording, each sample point consists of a 12 * 2-byte block, with leads ordered as I, II, III, aVR, aVL, aVF, V1, V2, V3, V4, V5, and V6 [4].

⚙️ Converting to a Voltage (The Key Step)

The 16-bit integer is just a "code" or "ADC unit." To make sense of it, you use the file's header information.

- **Amplitude Resolution:** The header contains an array of 12 integer values specifying the amplitude resolution for each lead, stored in **nanovolts (nV)** per ADC unit [4].
- **The Formula:** To convert the raw integer sample to a voltage, use this formula:
  `Voltage (in µV) = Raw_Integer_Value * Amplitude_Resolution (in nV) / 1000` [4]
- **Dynamic Range:** Because the integer has a fixed 16-bit range, the overall dynamic range (e.g., ±81.9 mV) is determined by the amplitude resolution. A less sensitive system (with a larger resolution, like 2500 nV) will have a larger dynamic range than a more sensitive one (with a smaller resolution) [4].

📝 Summary

So, to directly answer your question: the data is stored as a **16-bit signed integer**. It is only by reading the header's **amplitude resolution** field that you can interpret that integer as a float representing a voltage.

I hope this clarifies the structure for you. If you have more questions about a specific file or parser, feel free to ask

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> I am bringing this up because I need to figure out how to record ECG data in the ISHNE format from a system that reads the raw data from a ADS1115 ADC chip that captures data from a AD8232 heart monitor, BUT the raw data must be converted to float for the post processing of the ECG QRS detection (Pan-Tompkins), so the final data from that step is floating point - how do I then out the acceptable data to the file?

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To solve this, you'll need to save the 16-bit integers that represent the original ADC readings into the ISHNE file, not the processed float voltages. This is because the ISHNE format specifically stores raw, unprocessed samples from the analog-to-digital converter (ADC) [8] [11].

The key is to ensure the **amplitude resolution** value in your file's header matches the conversion you performed to get from the raw integer to the final float voltage. Here is a simple two-step process to follow:

🧮 Step 1: Calculate the Amplitude Resolution for Your File

For your setup (ADS1115 → AD8232 → Pan-Tompkins), you must determine the relationship between the raw integer and a physical voltage (in microvolts, µV) to set the correct ISHNE header.

1. **Find your ADC's resolution**: You can calculate this from your specific ADS1115 setup.
   - For example, at the default ±4.096V range with a 16-bit ADC, the resolution is `(4.096V * 2) / 65536 = 0.000125V = 125 µV` per ADC step.

- The raw integer from the ADS1115 (`ADC_Code`) represents `ADC_Code * 125 µV` .

2. **Calculate the amplitude resolution in nV**: The ISHNE header requires the amplitude resolution in **nanovolts (nV)** per ADC unit [5] [8] .
   - Using the example above, the amplitude resolution is `125 µV * 1000 nV/µV = 125,000 nV` .
   - This is the value you will store in the amplitude resolution field of the file header for each lead.

✍️ Step 2: Write the Data and Header

Now that you have your amplitude resolution, you can correctly store your data.

- **Writing the Data Block**: The ISHNE format stores the **raw** `ADC_Code` **as a 16-bit signed integer** [8] [11] . You cannot store the final processed float from your Pan-Tompkins algorithm. You must save the original, unscaled integer values from the ADS1115 (before any filtering or QRS detection) into the ISHNE file's data block.
- **Writing the Header**: In your file's header, you will set the "Amplitude Resolution" field (a 24-byte array of 12 values, one for each possible lead) to the value you calculated in **nV** (e.g., 125000) [5] [8] .

🔄 Reversing the Process

When a compliant ISHNE reader opens your file, it will use the amplitude resolution you provided to convert the raw integers back into a voltage. The relationship will be:
`Voltage (in µV) = Raw_Integer * (Amplitude_Resolution_nV / 1000)` .
This gives a float voltage, which can then be used for processes like QRS detection [3] [10] .

By storing the raw ADC counts and defining the correct amplitude resolution in the header, you ensure your file is valid and your data is correctly interpretable by any standard ISHNE reader.

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> One more thing: the sampling rate we are using is 215 Hz. A number of ECG examples use 200 Hz sampling rate. As long as

we output the sampling rate in the ISHNE file header, there should be no problems. Am I correct?

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Yes, you are absolutely correct. Specifying the sampling rate in the ISHNE file header is the standard and correct way to handle a custom sampling frequency, so your 215 Hz rate will be just fine.

📄 How the Sampling Rate is Handled in ISHNE

The ISHNE format is designed to be flexible. The sampling rate is not a fixed value; it is a defined parameter stored in the file's header.

- **The Header Field**: The specification includes a dedicated 2-byte field in the header specifically for the **"Sampling rate (in hertz)"** [1]. This is where you will store the value of `215`.
- **Why This Works**: Any standard software or library that reads an ISHNE file knows to look at this header value to correctly interpret the time intervals between samples. By setting this field correctly, you ensure that the 12-bit samples stored in the data block are properly associated with a time axis [7] [11].

⚖️ A Note on Your File's Size

The main practical implication of changing the sampling rate is the size of the data file. A higher sampling rate means more samples are recorded per second, which will create a larger file for the same recording duration. This is an expected trade-off for the benefit of a higher temporal resolution in your recording and is handled by the format without issue.

In short, you can confidently use your 215 Hz sampling rate as long as you accurately write that value into the header.

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